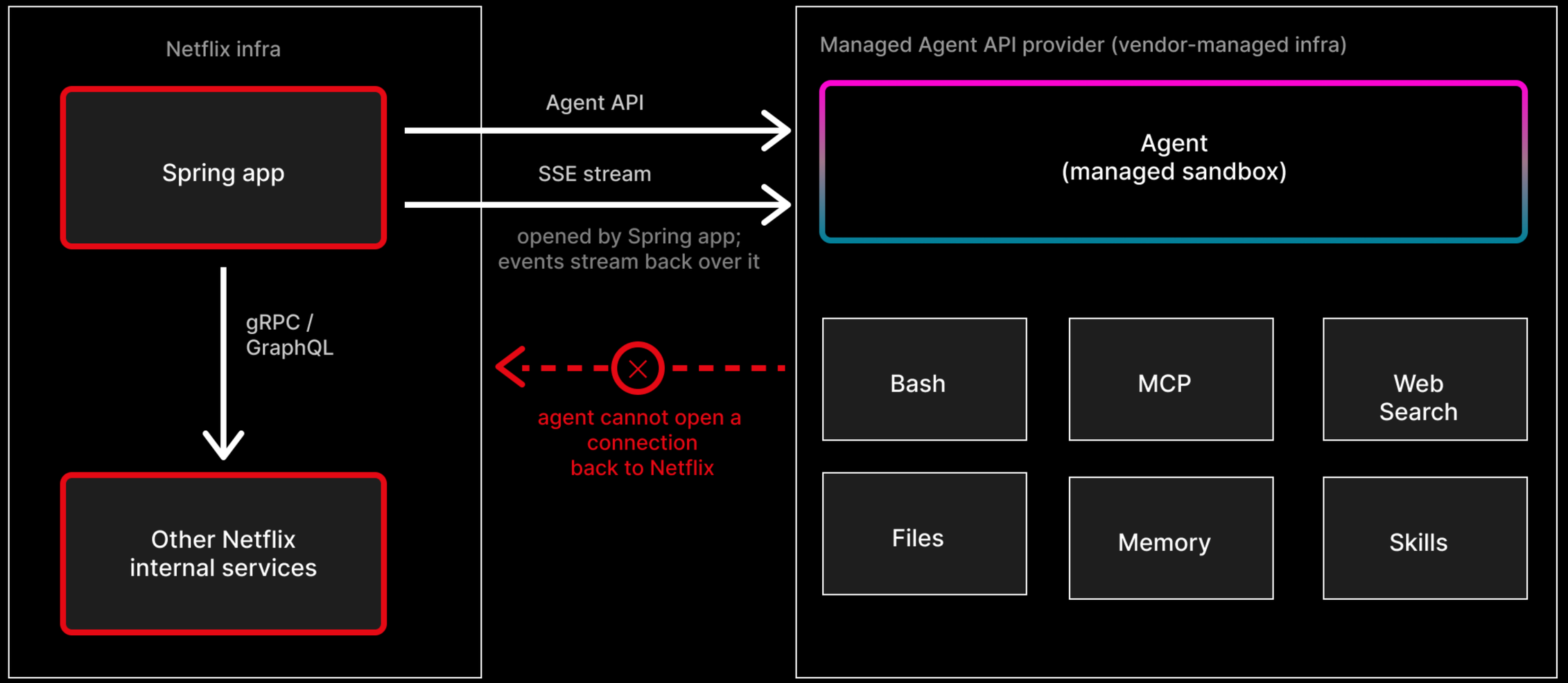
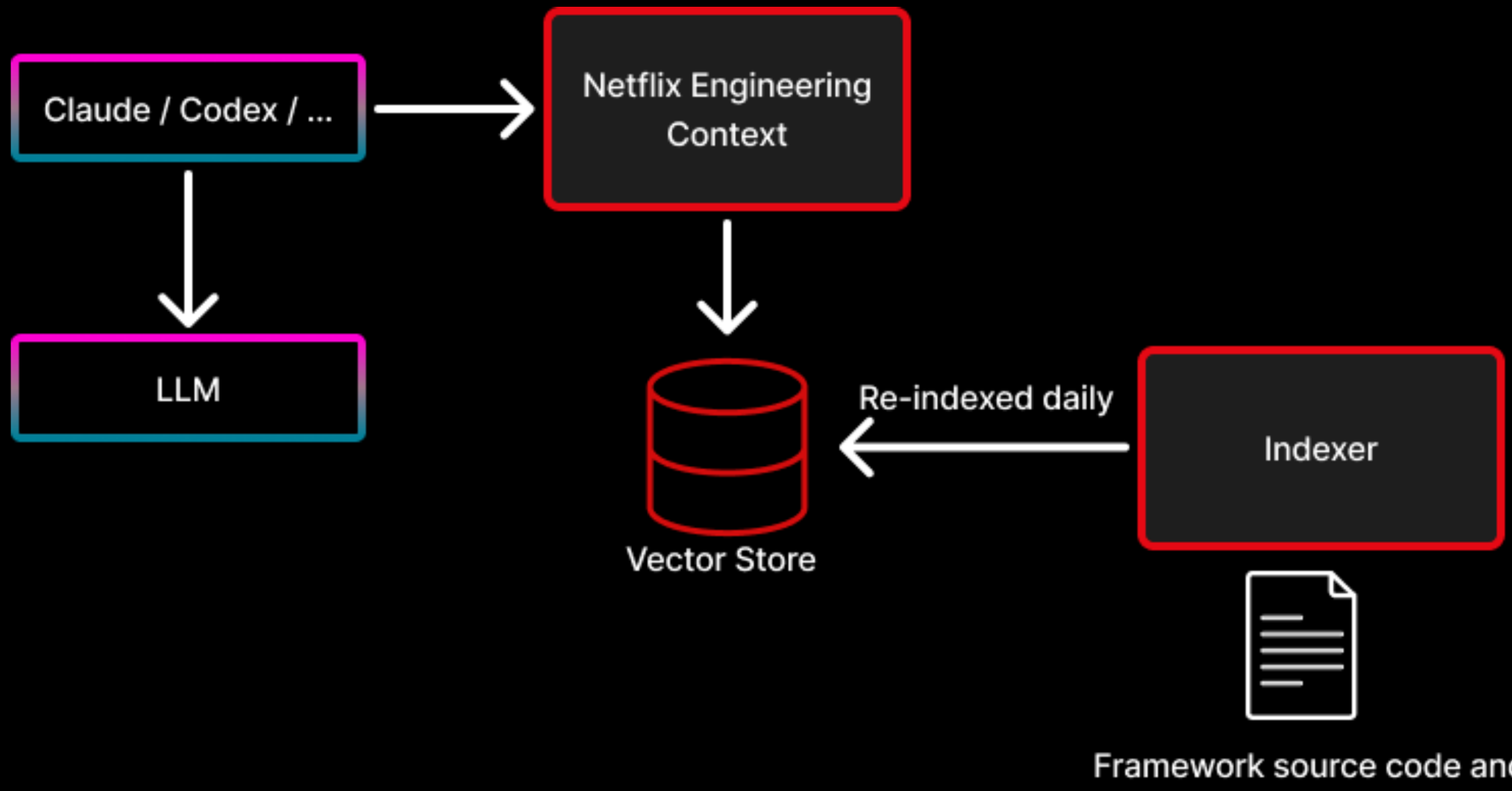
Managed Agent APIs



Coding agents



RAG for internal frameworks



Agents are bad at Java libraries...

- The project only uses `DgsQueryExecutor` as an autowired field in two tests. It doesn't implement the interface, so the implementation is in the DGS library.

  In the open-source DGS framework, `DgsQueryExecutor` is an interface in `com.netflix.graphql.dgs`. Its implementation is `DefaultDgsQueryExecutor`, a Kotlin class that DGS auto-configuration registers as a Spring bean. That's from memory, not from this repo's dependencies. I haven't confirmed that your resolved DGS version matches, or that Netflix's internal DGS build doesn't add its own implementation.

This class was removed in 2024 🫠

Agents are bad at working with JARs

```
❯ What is the implementation of DgsQueryExecutor
  ⎿ Error: No such tool available: Grep. Grep is not available in this session — search file contents with `grep` via the Bash tool instead.

● Bash(grep -rl "DgsQueryExecutor" --include='*.java' --include='*.kt' . 2>/dev/null | head -20)
  ⎿ sbn-dev-metrics-server/src/test/java/com/netflix/sbndevmetrics/datafetchers/AnalysisQueueDataFetcherTest.java
    sbn-dev-metrics-server/src/test/java/com/netflix/sbndevmetrics/datafetchers/StartupEventsDataFetcherTest.java
  ⎿ Allowed by auto mode classifier
  ⎿ 2 PostToolUse hooks ran

● Bash(find ~/.gradle ~/.m2 -name 'graphql-dgs-*.jar' 2>/dev/null | grep -v -- '-sources' | head -5; find ~/.gradle ~/.m2 -name 'graphql-dgs-[0-9]*-sources.jar' 2>/dev/null | head -3)
  ⎿ /Users/pbakker/.m2/repository/com/netflix/graphql/dgs/graphql-dgs-pacs-authz/11.0.92-snapshot.202605191823.uncommitted+dceb547/graphql-dgs-pacs-authz-11.0.92-snapshot.202605191823.uncommitted+dceb547.ja
    r
    /Users/pbakker/.m2/repository/com/netflix/graphql/dgs/graphql-dgs-pacs-authz/11.0.92-snapshot.202605192151.uncommitted+dceb547/graphql-dgs-pacs-authz-11.0.92-snapshot.202605192151.uncommitted+dceb547.ja
    r
    /Users/pbakker/.m2/repository/com/netflix/graphql/dgs/graphql-dgs-example-shared/10.6.0-SNAPSHOT/graphql-dgs-example-shared-10.6.0-SNAPSHOT-javadoc.jar
    /Users/pbakker/.m2/repository/com/netflix/graphql/dgs/graphql-dgs-example-shared/10.6.0-SNAPSHOT/graphql-dgs-example-shared-10.6.0-SNAPSHOT.jar
    /Users/pbakker/.m2/repository/com/netflix/graphql/dgs/graphql-dgs-netflix-authz-instrumentation/11.0.92-snapshot.202605191823.uncommitted+dceb547/graphql-dgs-netflix-authz-instrumentation-11.0.92-snapsh
    ot.202605191823.uncommitted+dceb547.jar
    /Users/pbakker/.gradle/caches/modules-2/files-2.1/com.netflix.graphql.dgs/graphql-dgs/12.1.0/77afc1bdd0c0edd1066dd40c6d1a3ed7ad947365/graphql-dgs-12.1.0-sources.jar
    /Users/pbakker/.gradle/caches/modules-2/files-2.1/com.netflix.graphql.dgs/graphql-dgs/12.0.0/77afc1bdd0c0edd1066dd40c6d1a3ed7ad947365/graphql-dgs-12.0.0-sources.jar
    /Users/pbakker/.gradle/caches/modules-2/files-2.1/com.netflix.graphql.dgs/graphql-dgs/12.0.1/77afc1bdd0c0edd1066dd40c6d1a3ed7ad947365/graphql-dgs-12.0.1-sources.jar
  ⎿ Allowed by auto mode classifier
  ⎿ 2 PostToolUse hooks ran

● Bash(cd /tmp && J=/Users/pbakker/.gradle/caches/modules-2/files-2.1/com.netflix.graphql.dgs/graphql-dgs/12.1.0/77afc1bdd0c0edd1066dd40c6d1a3ed7ad947365/graphql-dgs-12.1.0-sources.jar; unzip -l $J | grep -i
    'QueryExecutor')
  ⎿      8272  02-01-1980 00:00   main/com/netflix/graphql/dgs/internal/BaseDgsQueryExecutor.kt
         1176  02-01-1980 00:00   main/com/netflix/graphql/dgs/internal/DgsQueryExecutorRequestCustomizer.kt
        19777  02-01-1980 00:00   main/com/netflix/graphql/dgs/DgsQueryExecutor.java
  ⎿ Shell cwd was reset to /Users/pbakker/repo/sprng-sbn-dev-metrics
  ⎿ Allowed by auto mode classifier
  ⎿ 2 PostToolUse hooks ran

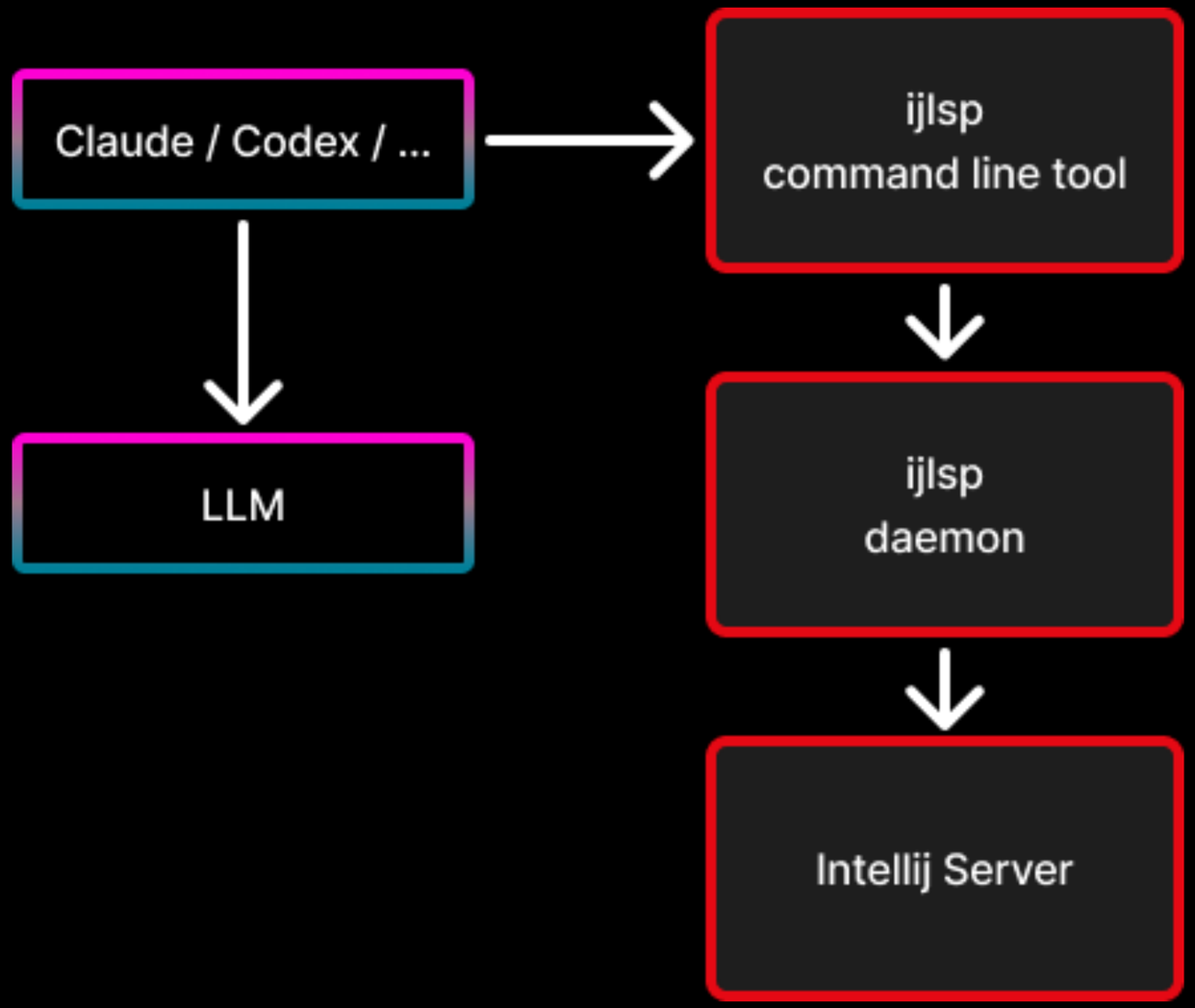
● Bash(cd /tmp && for J in $(find ~/.gradle/caches/modules-2 -path '*com.netflix.graphql.dgs*' -name '*12.1.0*-sources.jar' | head -40); do unzip -l $J | grep -E 'DefaultDgsQueryExecutor|:
    *DgsQueryExecutor|DgsQueryExecutor *\{' | sed "s|^|$(basename $J): |"; done)
  ⎿ Shell cwd was reset to /Users/pbakker/repo/sprng-sbn-dev-metrics
  ⎿ Allowed by auto mode classifier
  ⎿ 2 PostToolUse hooks ran

                                                                                                                                                    04:03 PM claude-sonnet-5-5
● Still looking: DgsQueryExecutor is an interface in the DGS jar, so I'm searching the sources jars for the class that implements it.

● Bash(cd /tmp && for J in $(find ~/.gradle/caches/modules-2 -path '*com.netflix.graphql.dgs*' -name '*-sources.jar' | grep -E '12\.1\.0' | head -60); do unzip -l $J 2>/dev/null | grep -E 'QueryExecutor' | sed
    "s|^|$(basename $J): |"; done)
  ⎿ graphql-dgs-spring-graphql-12.1.0-sources.jar:      7148  02-01-1980 00:00   main/com/netflix/graphql/dgs/springgraphql/SpringGraphQLDgsQueryExecutor.kt
```

* Baked for 1m 39s

Jetbrains LSP



Jetbrains LSP

```
● DgsQueryExecutor is an interface in graphql-dgs:12.0.1. Its implementation is SpringGraphQLDgsQueryExecutor in graphql-dgs-spring-graphql:12.0.1. This is SpringGraphQLDgsQueryExecutor.kt:41.

  ijlsp impls found no implementations, probably because the class is Kotlin. The api output gave its signature but not the method bodies, which I haven't read.

  Constructor dependencies:
  - ExecutionGraphQlService
  - DefaultDgsGraphQLContextBuilder
  - DgsDataLoaderProvider
  - DgsJsonMapper
  - DgsQueryExecutorRequestCustomizer
  - List<GraphQLContextContributor>

  Methods:
  - execute(query, variables, extensions, headers, operationName, webRequest) returns an ExecutionResult.
  - The executeAndExtractJsonPath overloads take variables, headers or a ServletWebRequest.
  - executeAndGetDocumentContext has two overloads.
  - The executeAndExtractJsonPathAsObject overloads take a Class<T> or a TypeRef<T>.

  DgsSpringGraphQLAutoConfiguration.springGraphQLDgsQueryExecutor (line 652) creates the bean. AnalysisQueueDataFetcherTest and StartupEventsDataFetcherTest in this repo autowire it.

  There is also a reactive variant, SpringGraphQLDgsReactiveQueryExecutor, which implements DgsReactiveQueryExecutor.

  I can read the method bodies, such as how execute builds the request and context, if you want them.

✻ Cogitated for 22s · done 4:35 PM
```




MCP vs GraphQL

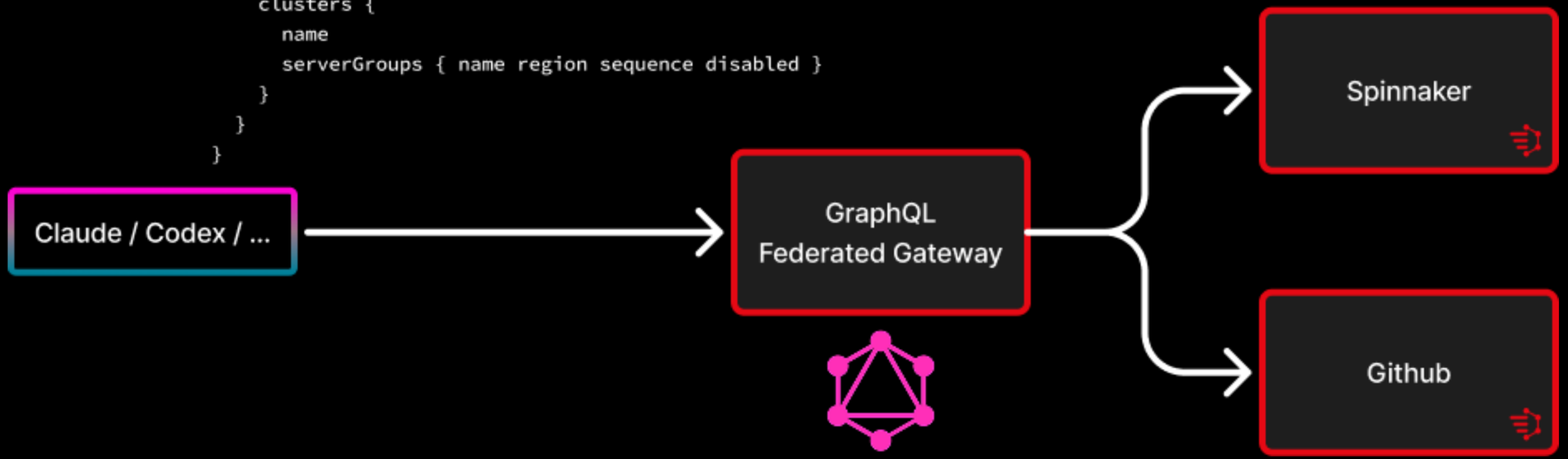
- MCP adds a lot of context overhead
- Most tools are not called every sessions
- Netflix is already standardized on Federated GraphQL for most systems

MCP vs GraphQL

- Should we use GraphQL APIs instead of MCP?
- Agents are good at discovering queries from a schema

MCP vs GraphQL

```
query SbnDevMetrics {
  spin_applicationsByIds(ids: ["sbndevmetrics"]) {
    metadata {
      sourceRepository {
        url
        defaultBranch { name }
      }
    }
    clusters {
      name
      serverGroups { name region sequence disabled }
    }
  }
}
```



Java vs Python for GenAI

- ~2 years ago, most GenAI libraries where Python
- Java has caught up with Spring AI etc.



Java vs Python for GenAI

So it’s back to the old question:

“What platform are you most skilled in, and how fast does it need to run?”

“You’re absolutely right. I should just use Java”
- Some Agent

THANKS